Outline of Goals for May-June 2026

Max Watzky*

May 18th, 2026

1. **Week 1 (May 18th - 22nd)** My primary goal for the first week of the summer is to wrap up the projects and analyses from the preceeding semester. I will begin by cleaning up and documenting the utility codes (the Richards-Wolf integration, plotting tools, angular spectrum propagation etc.) that I have written over the past year, and organizing them in a GitHub repository. Then, I will finish the process of rigorously verifying the correctness of the Richards-Wolf code. On Wednesday, Shengqi and I will take images of a bead under various aberrations, and the resulting data will be compared to the simulated PSFs. This was attempted back in April, but insufficient documentation of the imaging parameters prevented us from drawing definitive conclusions. If any errors are discovered, the rest of the week will be spent attempting to correct them, or repeating the imaging session if necessary. Otherwise, I will begin the process of organizing the data from the Booth algorithm experiments we conducted over February-March. **By the end of the week, I hope to complete a brief writeup or slideshow with figures demonstrating the correctness of both the integration and correction codes.**

2. **Week 2 (May 25th - 29th)** This week, I will concentrate on literature review, with a focus on advances in AI/ML methods in modal AO. Traditionally, this means taking an aberrated image and directly inferring the Zernike coefficients in a single shot. However, preliminary experiments with the Booth algorithm have revealed that even the most robust tools can fall into the "traps" of local minima in the optimization landscape. I hypothesize that most of the current machine learning methods for modal AO are susceptible to this inefficiency, and my literature review will focus on rigorously (and impartially) evaluating this hypothesis.

3. **Week 3, 4, 5 (June 1st - 19th)** Building on the results from last week, I hope to replicate the results of a recent AI/ML paper for modal AO, and demonstrate the local minima "trap" directly. Ideally, this will only take one or two weeks of work. However, if the authors of the paper have not published theri code, it may take substantially longer.

   Looking towards the Fall semester, I will also begin reading up on probabilistic methods, such as diffusion. More than merely predicting the aberration, probabilistic tools have the capability to generate an entire distribution of possible aberrations. This is useful because it is relatively easy to "check" an aberration by applying its sign inverse to the SLM and measuring the quality of the correction. By iteratively sampling from the probability distribution of aberrations in this manner, the ideal correction will be identified most often, mitigating the chance of falling into a local minimum.

*Departments of Physics and Mathematics, Yale University